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PRESENTATIONS & APPEARANCES

I have given numerous presentations on the work being done by me and my collaborators to the scientific community and the public along with providing lectures and tutorials and participating in panel discussions. The list below is mostly complete beyond 2022 (the beginning of my tenure as a faculty member) and modestly complete before then (2011-2022).

Invited Talks

AstroAI Summer Workshop	Jun 2024
AI SpAI with My (Little?) AI: Applications of Statistical Learning in Modern Astrophysics	
NSF IAIFI Summer Workshop	Aug 2023
Uncertainty and Interpretability in Machine Learning Models	
MaxEnt 2023 Workshop	Jul 2023
Frontiers of Nested Sampling Meeting-within-a-Meeting	
Musings on Nested Sampling from Interactions with Everyday Users	
Statistical Challenges in Modern Astronomy VIII	Jun 2023
Extending Bayesian Modeling Approaches to ML Applications	
IAU General Assembly 2022	Aug 2022
Machine Learning in Astronomy: Possibilities and Pitfall	
Incorporating Errors into Machine Learning Methods	
UCLA IPAM	Nov 2021
Mathematical and Computational Challenges in the Era of Gravitational Wave Astronomy	
An Introduction to (Dynamic) Nested Sampling	
University of Surrey	Jul 2021
Towards a Cross-Research Platform for Hosting Bayesian Data-Fitting Tools	
Keynote Address: An Introduction to Nested Sampling	
Harvard University	Aug 2018
CMSA Big Data Conference	
Revealing the Milky Way's Dust-iny	

Colloquia & Seminars

Tsinghua University Colloquium “Trust Me, I am an Astrophysics Foundation Model”	Dec 2024
Ohio State University CCAPP Seminar I Spy with my (Little?) AI: Applications of Statistical Learning in Modern Astrophysics	Mar 2024
University of Guelph Math & Stats Seminar Nested Sampling: Past, Present, & Future	Oct 2023
Université de Montréal Astrophysics Seminar Astrostatistical Thinking in Machine Learning Applications	Oct 2023
Carnegie Mellon University STAMPS Seminar Nested Sampling: Past, Present, and Future	Sep 2023
University of Pittsburgh Astronomy Seminar From Panchromatic Images to Stellar Ages and Beyond	Sep 2023
Center for Astrophysics Harvard & Smithsonian AstroAI Seminar Extending Statistical Thinking to ML Applications	Aug 2023
McMaster University Astronomy Lunch Seminar From Panchromatic Images to Stellar Ages and Beyond	Apr 2023
University of Waterloo Astroseminar From Panchromatic Images to Stellar Ages and Beyond	Feb 2023
NRC Herzberg Astronomy and Astrophysics Research Centre Colloquium Galaxy Evolution through the Eyes of the Milky Way	Nov 2022
Queen’s University Astronomy Seminar Galaxy Evolution through the Eyes of the Milky Way	Nov 2022
Institute for Advanced Study	Aug 2022

Astrophysics Seminar From Panchromatic Images to Stellar Ages and Beyond	
Saint Mary's University Department of Astronomy and Physics Colloquium More Data + More Parameters = More Fun with Galaxy Evolution	Aug 2022
CASCA CANadian Virtual Astronomy Seminar (CANVAS) Mapping the Milky Way Near and Far	Apr 2022
IAU-IAA Astrostats and Astroinfo Seminar Statistical Challenges in Stellar Parameter Estimation from Theory and Data	Apr 2022
Saint Mary's University Data Analytics Seminar An Introduction to (Dynamic) Nested Sampling	Jan 2022
University of Toronto Toronto Data Workshop Stars, Galaxies, and Everything In-Between: Galaxy Evolution Near and Far with Large Datasets	Oct 2021
Penn State University Department of Astronomy and Astrophysics Colloquium Mapping the Milky Way with Stars and Dust	Sep 2021
University of Chicago Kavli Institute for Cosmological Physics Seminar Cosmological Cartography with Photometric Redshifts	Apr 2021
CANSSI Ontario Data Science Applied Research and Education Seminar Mapping the Milky Way in the Age of Gaia	Feb 2021
University of Florida Department of Astronomy Colloquium Enabling Data-Driven Discovery in the Milky Way and Beyond Using Large Astronomical Datasets	Sep 2020
Villanova University Department of Astrophysics and Planetary Science Colloquium Exploring the Galaxy Near and Far in the Age of Gaia	Oct 2019
Harvard University Summer Colloquium (joint with Catherine Zucker)	Jun 2019

Charting Nearby Molecular Clouds with Gaia: A New Map of Our Local Interstellar Medium

University of Cambridge Apr 2019

Data Intensive Science Seminar

Mapping the 3-D Distribution of Dust in the Milky Way with Stellar Photometry

UMass Amherst Oct 2017

Data Science Tea

Big Data Inference: Using Hierarchical Bayes and Machine Learning to Improve Photometric Redshifts

Harvard University Sep 2017

CHASC Astrostatistics Seminar

An Introduction to Dynamic Nested Sampling

Kavli IPMU Mar 2016

Astronomy Lunch Seminar

Mapping, Visualizing, and Exploiting the Color-Redshift Relation

University of Tsukuba Aug 2013

Theoretical Astrophysics Seminar

The Evolution of Star-Forming Galaxies over Cosmic Time

Contributed Talks

Banff International Research Station Nov 2023

Astrostatistics in Canada and Beyond

SBI++: Extending Simulation-Based Inference to Censored and Out-of-Distribution Data

University of Michigan Aug 2023

2023 Clusters & Streams Workshop

Rapid Stellar Parameter Inference with Probabilistic Machine Learning

JSM 2023 Aug 2023

True North Strong and... Amazing at Astrostatistics! (Invited Paper Session)

Connecting Galaxy Evolution between Observation, Simulation, and Theory

Dunlap Institute for Astronomy & Astrophysics Jun 2023

15th Anniversary Homecoming Celebration Spotlight Talk

Uncovering Galaxy Evolution through the Eyes of the Milky Way at the Dunlap Institute

Surveys to Discovery (S2D) Conference May 2023

Photometric Biases in Modern Surveys

2023 Spring Research Conference (SRC) May 2023

From Light to Mass and Back Again: Exploring Galaxy Evolution with “New” Methods and “Old” Outcomes

McMaster University Aug 2022

2022 Clusters Workshop

Panchromatic Modelling of Co-Eval Stellar Populations

JSM 2022 Aug 2022

Advances in Astrostatistics in the Great White North (Invited Paper Session)

Mapping the Milky Way in 5-D with Big Data

Seeing the Future: A Conference in Honour of Alyssa Goodman May 2022

Machine Learning in the Era of Astronomically Big Data

AAS 238 Jun 2021

Special Session (Statistics Discussant)

Unaccounted Uncertainties: The Role of Systematics in Astrophysics

JSM 2020 Aug 2020

Astrostatistics Interest Group: Student Paper Award (Topic-Contributed Paper Session)

Photometric Biases in Modern Astronomical Surveys

Lorentz Center: Colours of the Universe Workshop Sep 2018

Challenges Working with Posterior Distributions (with Alex Malz)

COSMO21 2016 May 2016

Improving Photometric Redshifts for Hyper Suprime-Cam

Lectures & Tutorials

Summer School in Statistics for Astronomers Jun 2024

Nested Sampling: Past, Present, and Future

2023 ESO Conference Sep 2023

Two in a million – The interplay between binaries and star clusters

Machine Learning for Star Cluster Science (with Steffani Grondin)

University of Michigan Aug 2023

2023 Clusters & Streams Workshop

Dimensionality Reduction: Applications to Star Cluster Science (with Steffani Grondin)

NSF IAIFI Summer School Aug 2023

Uncertainty and Interpretability in Machine Learning Models (with Alex Gagliano)

Statistical Challenges in Modern Astronomy VIII Jun 2023

A Tutorial of (Dynamic) Nested Sampling

Astro Hack Week 2020 Aug 2020

Introduction to Bayesian Inference with Linear Regression

Panels

NeurIPS 2023	Dec 2023
Workshop on Machine Learning and the Physical Sciences	
Panel on inductive biases and interpretability	
NSF IAIFI Summer School	Aug 2023
Career Panel	
Dunlap Institute for Astronomy & Astrophysics	Jun 2023
15 th Anniversary Homecoming Celebration	
From Dunlap to Diverse Careers: A Panel Discussion on Opportunities and Challenges for Astrophysics Graduates (Moderator)	

Posters

2023 ESO Conference	Sep 2023
Two in a million – The interplay between binaries and star clusters	
A machine-learning led search for extra-tidal stars of globular clusters	
2022 CASCA Annual General Meeting	May 2022
Data Driven Stellar Spectral Modelling with GSPICE	
2021 CASCA Annual General Meeting	May 2021
Deriving Stellar Properties, Distances, and Reddenings from Photometry (and Astrometry) with brutus	
Bayes Comp 2018	Mar 2018
Dynamic Nested Sampling with <i>dynesty</i>	
AAS 229	Jan 2017
Improving Photometric Redshifts for Hyper Suprime-Cam (HSC) with Hierarchical Bayes and Machine Learning	
AAS 223	Jan 2014
Parallel Galaxy Main Sequence and Quasar Evolution from $z=0-6$ (AAS Chamblis Award Honorable Mention)	
AAS 221	Jan 2013
An In-Depth Analysis of the Kepler Low-Amplitude Blazhko RR Lyrae Stars: First Steps and Comparing Data Reduction Methods	
2012 KASC Meeting	Jun 2012
An In-Depth Analysis of the Kepler Low-Amplitude Blazhko RR Lyrae Stars: First Steps and Comparing Data Reduction Methods	

AAS 219	Jan 2012
The X-ray Counterpart of the High-B Pulsar J0726-2612 (AAS Chamblis Award)	

Public Talks & Events

Astronomy on Tap (Toronto)	Jul 2024
Host	
Christie Gardens Apartments & Care (Toronto)	Nov 2023
The Role of AI in Modern Astrophysics	
Astronomy on Tap (Toronto)	Nov 2022
Mapping the Milky Way from the Inside Out	
RASC (Toronto Centre)	Oct 2021
Speaker's Night	
Mapping the Milky Way in the Age of Gaia	
RASC (Ottawa Centre)	Jun 2021
Monthly Meeting	
Mapping the Milky Way in the Age of Gaia	
GitHub Satellite 2019	May 2019
Keynote participant for open-source code contributions (<i>dynesty</i>) in the analysis of M87* by the EHT collaboration	

Media Appearances

CBC Calgary	Jan 2022
Calgary Eyeopener with David Gray, Angela Knight	
Radio Interview: "Space bubble" (link)	